ECET 250E-Linear Circuit 1
Problemset 6-Chapter 7- First Order RC-RL Circuits
Solution

Name: _____ Marks: _____

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1. **7.2** Find the time constant for the *RC* circuit in **Fig. 7.82**.

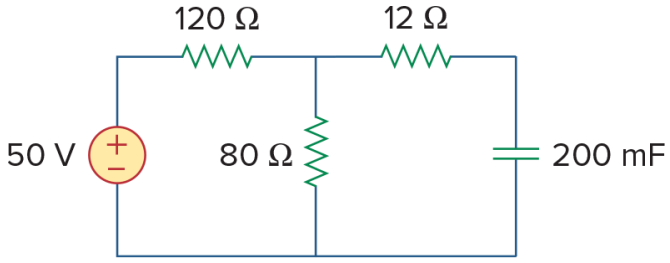


Figure 7.82 For **Prob. 7.2**.

Chapter 7, Solution 2.

$$\tau = R_{th}C$$

where R_{th} is the Thevenin equivalent at the capacitor terminals.

$$R_{th} = 120 \,||\, 80 + 12 = 60\ \Omega$$

$$\tau = 60 \times 200 \times 10^{-3} = \mathbf{12\ s}.$$

2. **7.6** The switch in **Fig. 7.86** has been closed for a long time, and it opens at t=0. Find v(t) for t≥0.

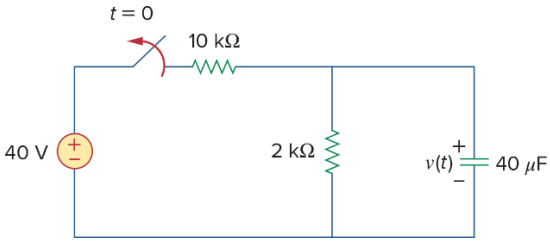


Figure 7.86 For **Prob. 7.6**.

Chapter 7, Solution 6.

$$v_o = v(0) = \frac{2}{10+2}(40) = 6.667\,V$$

$$v(t) = v_o e^{-t/\tau},\ \tau = RC = 40x10^{-6}\,x2x10^{3} = \frac{2}{25}$$

$$v(t) = \underline{6.667e^{-12.5t}\,V}$$

3. 7.9 The switch in **Fig. 7.89** opens at t=0. Find vo for t>0.

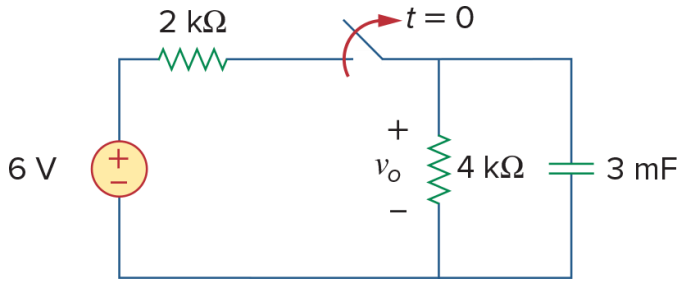


Figure 7.89 For **Prob. 7.9**.

Chapter 7, Solution 9

For t < 0, the switch is closed so that

$$v_o(0) = \frac{4}{2+4}(6) = 4\ \text{V}$$

For t >0, we have a source-free RC circuit.

$$\tau = RC = 3x10^{-3}x4x10^{3} = 12s$$

$$v_o(t) = v_o(0)e^{-t/\tau} = \mathbf{4e^{-t/12}\ V}.$$

4. 7.11 For the circuit in **Fig. 7.91**, find i_o for t>0.

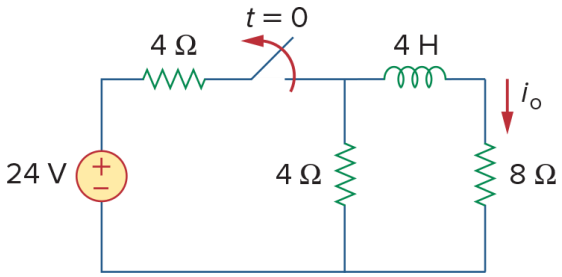
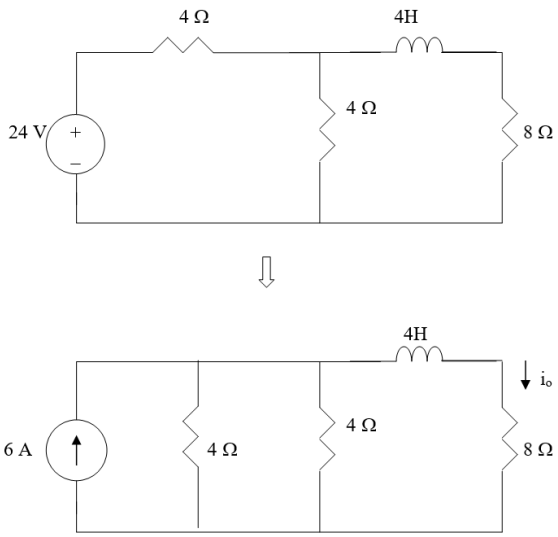


Figure 7.91 For **Prob. 7.11**.

Chapter 7, Solution 11.

For t<0, we have the circuit shown below.



4||4= 4x4/8=2

$i_o(0^-) = [2/(2+8)]6 = 1.2$ A

For t >0, we have a source-free RL circuit.

$\tau = \frac{L}{R} = \frac{4}{4+8} = 1/3$ thus,

$i_o(t) = \mathbf{1.2e^{-3t}\ A}$.

5. 7.13. In the circuit of **Fig. 7.93**,

$$v(t) = 80e^{-10^3 t}\ \text{V}, \quad t > 0$$
$$i(t) = 5e^{-10^3 t}\ \text{mA}, \quad t > 0$$

(a)Find *R*, *L*, and τ.

(b)Calculate the energy dissipated in the resistance for 0<t<0.5 ms.

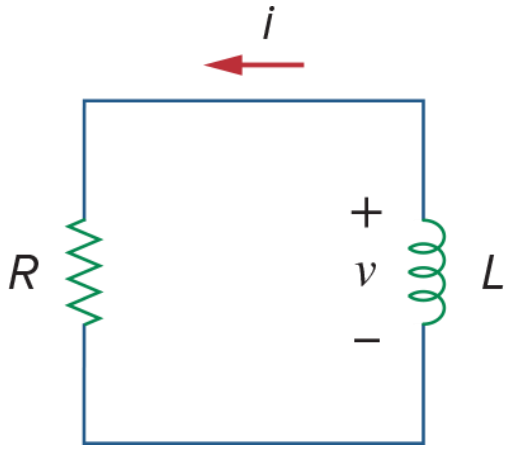


Figure 7.93 For **Prob. 7.13**.

Chapter 7, Solution 13.

(a) $\tau = \dfrac{1}{10^3} = \underline{1ms}$ = **1 ms**.

$v(t) = i(t)R = 80e^{-1000t}$ V = $R5e^{-1000t}x10^{-3}$ or R = 80,000/5 = **16 kΩ**.

But $\tau = L/R = 1/10^3$ or $L = 16x10^3/10^3$ = **16 H**.

(b) The energy dissipated in the resistor is

$$w = \int_0^{0.0005} pdt = \int_0^{0.0005} 0.4\, e^{-2000t} dt = -\frac{0.4}{2000} e^{-2000t}\Big|_0^{0.0005}$$

$$= 200(1-e^{-1})\ x10^{-6} = \textbf{126.42 µJ}.$$

(a) **16 kΩ, 16 H, 1 ms** **(b) 126.42 µJ**

6. 7.14 Calculate the time constant of the circuit in **Fig. 7.94**.

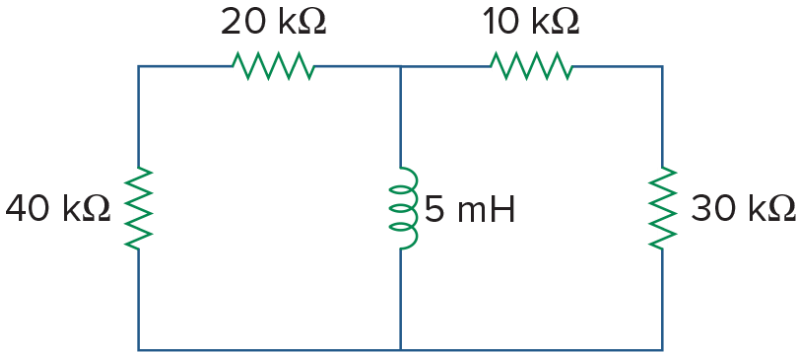


Figure 7.94 For **Prob. 7.14**.

Chapter 7, Solution 14.

$$R_{Th} = (40+20)//(10+30) = \frac{60x40}{100} = 24k\Omega$$

$$\tau = L/R = \frac{5x10^{-3}}{24x10^3} = \underline{0.2083\mu s}$$

7. **7.39** Calculate the capacitor voltage for t<0 and t>0 for each of the circuits in **Fig. 7.106**.

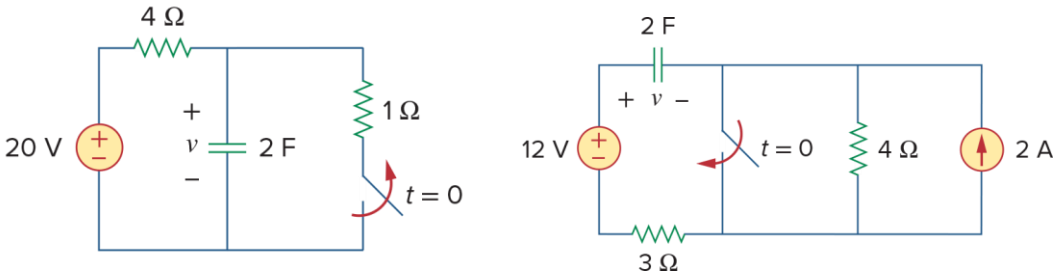


Figure 7.106 For **Prob. 7.39**.

Chapter 7, Solution 39.

(a) Before t = 0,

$$v(t) = \frac{1}{4+1}(20) = \mathbf{4\ V}$$

After t = 0,

$$v(t) = v(\infty) + \left[\, v(0) - v(\infty) \right] e^{-t/\tau}$$

$$\tau = RC = (4)(2) = 8, \quad v(0) = 4, \qquad v(\infty) = 20$$

$$v(t) = 20 + (4 - 20)\, e^{-t/8}$$

$$v(t) = 20 - 16\, e^{-t/8}\ \text{V}$$

(b) Before t = 0, $v = v_1 + v_2$, where v_1 is due to the 12-V source and v_2 is due to the 2-A source.

$$v_1 = 12\ \text{V}$$

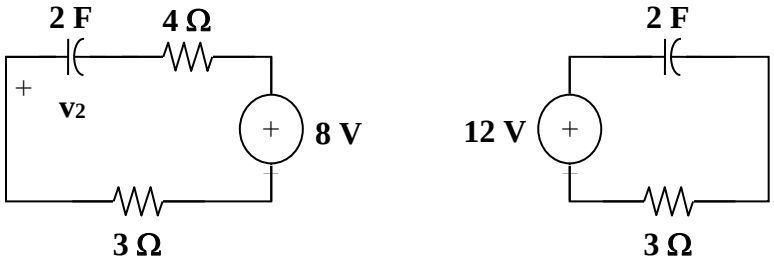
To get v_2, transform the current source as shown in Fig. (a).

$$v_2 = -8\ \text{V}$$

Thus,

$$v = 12 - 8 = \mathbf{4\ V}$$

After t = 0, the circuit becomes that shown in Fig. (b).



(a) **(b)**

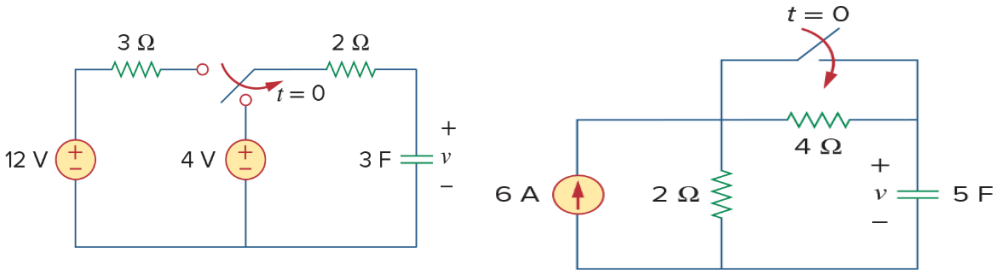
$$v(t) = v(\infty) + \left[v(0) - v(\infty) \right] e^{-t/\tau}$$

$$v(\infty) = 12, \qquad v(0) = 4, \qquad \tau = RC = (2)(3) = 6$$

$$v(t) = 12 + (4 - 12)\, e^{-t/6}$$

$$v(t) = \mathbf{12 - 8e^{-t/6}\ V}$$

8. 7.40 Find the capacitor voltage for t<0 and t>0 for each of the circuits in **Fig. 7.107**.



(a) (b)

Figure 7.107 For **Prob. 7.40**.

Chapter 7, Solution 40.

(a) Before t = 0, $v = \mathbf{12\ V}$.

After t = 0, $v(t) = v(\infty) + \left[v(0) - v(\infty) \right] e^{-t/\tau}$

$$v(\infty) = 4, \qquad v(0) = 12, \qquad \tau = RC = (2)(3) = 6$$

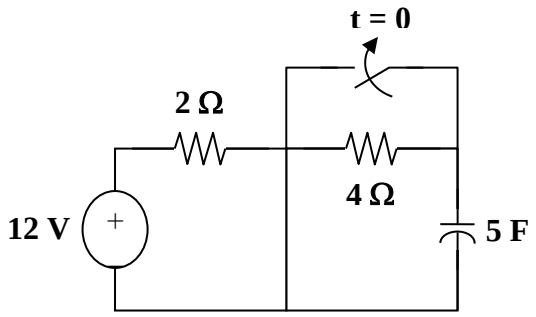
$$v(t) = 4 + (12 - 4)\, e^{-t/6}$$

$$v(t) = \mathbf{4 + 8e^{-t/6}\ V}$$

(b) Before t = 0, $v = \mathbf{12\ V}$.

(c) After t = 0, $v(t) = v(\infty) + \left[v(0) - v(\infty) \right] e^{-t/\tau}$

After transforming the current source, the circuit is shown below.



$v(0) = 12$, $v(\infty) = 12$, $\tau = RC = (2)(5) = 10$ $v = \mathbf{12\ V}$

9. 7.42 (a)If the switch in **Fig. 7.109** has been open for a long time and is closed at t=0, find $v_o(t)$.
(b)Suppose that the switch has been closed for a long time and is opened at t=0. Find $v_o(t)$.

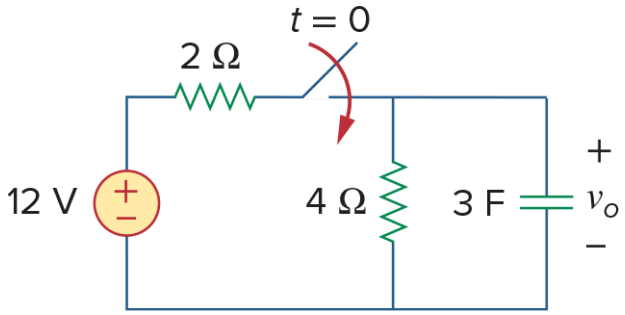


Figure 7.109 For **Prob. 7.42**.

Chapter 7, Solution 42.

(a) $v_o(t) = v_o(\infty) + \left[v_o(0) - v_o(\infty) \right] e^{-t/\tau}$

$$v_o(0) = 0,\ v_o(\infty) = \frac{4}{4+2}(12) = 8$$

$$\tau = R_{eq} C_{eq},\qquad R_{eq} = 2 \,||\, 4 = \frac{4}{3}$$

$$\tau = \frac{4}{3}(3) = 4$$

$$v_o(t) = 8 - 8e^{-t/4}$$

$$v_o(t) = \mathbf{8(1-e^{-0.25t})\ V}$$

(b) For this case, $v_o(\infty) = 0$ so that

$$v_o(t) = v_o(0)\, e^{-t/\tau}$$

$$v_o(0) = \frac{4}{4+2}(12) = 8, \qquad \tau = RC = (4)(3) = 12$$

$$v_o(t) = \mathbf{8e^{-t/12}\ V}$$